SEUNGMAN CHOI

Ph.D. Candidate, Purdue University
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EDUCATION & EXPERIENCES

2022.08–Present Ph.D., Purdue University, USA
- Advisor: Prof. Zubin Jacob
- Projects: Super-resolution imaging [6, 9–12, 15], and computational imaging for detection and ranging [14].

2019.04–2022.08 Research Associate (military duty), KAIST, KR
- Advisors: Prof. Seung-Woo Kim and Prof. Young-Jin Kim
- Project: Controllable EUV/DUV sources for semiconductor inspection [3–5,8]

2017.04–2019.03 M.S., Tokyo Institute of Technology, JP
- Advisor: Prof. Takasi Nisisako
- Project: Scaled-up microfluidic device [7, 13]

2013.04–2017.03 B.S., Tokyo Institute of Technology, JP (Magna Cum Laude)
- Advisor: Prof. Shigeki Saito
- Project: Electrostatic gripping system for nonplanar components [1, 2]

HONORS, AWARDS & GRANTS

2027.01–2027.05 Bilsland Fellowship (Pre-selected), $31,000 for a spring semester, Purdue University, USA

2026.01 CIDAR Semifinal Winner, DARPA, $50,000 awarded to the Zetascope team; core member, USA

2025.02 Outstanding Presentation Award, 10th 2025NCC, KR

2024.11 Outstanding Presentation Award, Autumn 2024KSPE, KR

2023.05 Best Presentation Award, 18th 2023IEEE-NEMS, KR

2022.11 Best Paper Award, 9th 2022ASPEN, SG

2022.11 Bronze Prize, Autumn 2022KSME, KR

2021.11 Best Paper Award, Next Generation Lithography Conference (2021OSK), KR

2019.01 Excellence Presentation Award, 2019KSEAJ, JP

2017.04–2019.03 ITO Educational Foundation Scholarship, ITO Found, JPY 180,000 per month, JP

2018.08 Foreign Tour Business Grant, PMTP-F, JPY 197,000 once, JP

2017.03 Muto Eiji Academic Excellence Award, JSDE, JP

2013.04–2017.03 Korean–Japan Joint Government Scholarship, JPY 130,000 per month, KR&JP

PUBLICATIONS

- **PREPRINTS & MANUSCPIPTS**

[15] **S. Choi**, et al., "Decoding atmospheric turbulence for few-frame diffraction-limited imaging", In preparation.
[14] S. Jape, **S. Choi**, et al., "Night vision transformers: hyperspectral reconstruction and passive depth estimation from a single broadband thermal image", Under review at *J. Opt. Soc. Am. A.*
[13] T. Nisisako, **S. Choi**, et al., "Mechanically and directionally tunable step emulsification via in-plane stretching of an elastomeric microfluidic device", Under review at *Ind. Eng. Chem. Res*.
[12] **S. Choi**, et al. "Adaptive information mining: Approaching the quantum limit of super-resolution imaging", Under review at *Optica* (arxiv.org/abs/2610.37491).
[11] P. Menart†, **S. Choi†**, et al., "Definition of multi-parameter super-resolution for optical telescopes", Under review at *Optics Express (*arxiv.org/abs/2610.14286).
[10] **S. Choi**, et al., "Photon-starved imaging through turbulence at the diffraction limit", Under review at *Laser & Photonics Reviews* (arxiv.org/abs/2510.22806).

- **PUBLISHED ARTICLES**

[9] **S. Choi**, et al., "Poisson wavefront imaging in photon-starved scenarios", *J. Opt. Soc. Am. B*, 43, 231-238, 2026.
[8] S. Won, J. Kim, Y. Oh, T. Kim, **S. Choi**, et al., "Programmable structured DUV illumination by coherent harmonic generation at crystalline solids for nanometer-resolution inspection of periodic samples", *PhotoniX*, *6*(1), 1-15, 2025.
[7] N. Tottori, **S. Choi**, et al., "Production of monodisperse oil-in-water droplets and polymeric microspheres below 20 μm using a PDMS-based step emulsification device", *Micromachines*, *16*(2), 132, 2025.
[6] H. Choi†, **S. Choi†**, et al., "Telescope imaging beyond the Rayleigh limit in extremely low SNR", New Journal of Physics 26 (9), 093019, 2024.
[5] B. Kim†, **S. Choi†**, et al., "Coherent manipulation of extreme-ultraviolet Bessel vortex beam from solids by active wavefront shaping of driving fundamental beam", *ACS Photonics*, 10.10: 3458-3466, 2023.
[4] S. Won†, **S. Choi†**, et al., "Dynamic ultraviolet harmonic beam pattern control by programmable spatial wavefront modulation of near-infrared fundamental beam" *Nanophotonics*, 12 16, 2023.
[3] Y. Kim†, B. Kim†, **S. Choi†**, et al., "Compensation of laser propagation effects within solids for high harmonic generation of extreme ultraviolet radiation" *Opt. Laser Technol*., 148 107507, 2022.
[2] Y. Taoka, K. Kawabata, P. Hemthavy, **S. Choi**, et al., "Development of Bipolar Electrostatic Chuck with a Beam-Array Assembly Fabricated by Lithography". *International Journal of Automation Technology*, *16*(4), 471-477, 2022.
[1] **S. Choi**, et al., "Development of a bipolar electrostatic chuck module with a beam-array assembly using the multiple etching process", *J. Micromech. Microeng.* 28 125011, 2018,
† Equal contribution.

PATENTS APPLICATIONS

- Z. Jacob, **S. Choi**, and P. Menart, **"**Optical System with Spatiotemporally Varying Point Spread Function for Image Super-resolution", US App. No. 19/847,161, 2026.
- Z. Jacob, **S. Choi**, and P. Menart, "System and method for photon-starved imaging through turbulence at the diffraction limit", US App. No. 64/078,003, 2026.
- S.-W. Kim, B. Kim, **S. Choi**, Y.-J. Kim, Y.-W. Kim, and S. Won "Apparatus and Method for Extreme Ultraviolet Beam Generation", KR Patent No. 10-2968825, 2026.
- T. Nisisako, **S. Choi**, and N. Tottori, "Micro Step-emulcification Device", JP App. No. 2018-036016, 2018.

RECENT RESEARCH PROJECTS

- **Diffraction-limited On-sky Imaging through Turbulence at McDonald Observatory (2025.11.24–12.03, 9 nights):** On-sky observations to demonstrate rapid, few-frame diffraction-limited imaging through atmospheric turbulence.
- **Super-resolution On-sky Imaging for Exoplanet Survey at WIYN Observatory (2026.05.20–22; Director's discretionary time, 1 night):** Initial on-sky observations to demonstrate sub-diffraction-limit resolution in binary-star imaging.
- **Computational Imaging for Detection and Ranging (Dec. 2024–May. 2026):** Analysis of DARPA-provided datasets to validate passive ranging over kilometer-scale distances without active illumination.
- **Polarization-resolved Computational Imaging for Semiconductor Inspection (Ongoing):** Developing defect detection methods by analyzing polarization-dependent responses of local electromagnetic modes in subwavelength semiconductor structures.

SKILLS

Programming languages	Python, MATLAB, and LabVIEW
Optical system design	Vacuum-based optical systems, microscopy, and astronomical instrumentation.
Computational methods	Coherent diffraction imaging (ptychography), blind deconvolution, deep image prior, and genetic algorithm.
Theory & Modeling	Estimation theory, information theory, quantum parameter estimation, and statistical noise modeling
Clean room nanofabrication	Deep reactive-ion etching and photolithography